



DATA WAREHOUSE

project

ABSTRACT

This report presents a Data Warehouse system designed to integrate and transform data from multiple sources (CRM and ERP) into a centralized repository. Using a layered architecture (Bronze, Silver, Gold) and ETL processes, raw data is cleaned, structured, and prepared for analysis. The final model uses a Star Schema to support efficient reporting and business intelligence. This solution improves data quality, consistency, and decision-making capabilities.

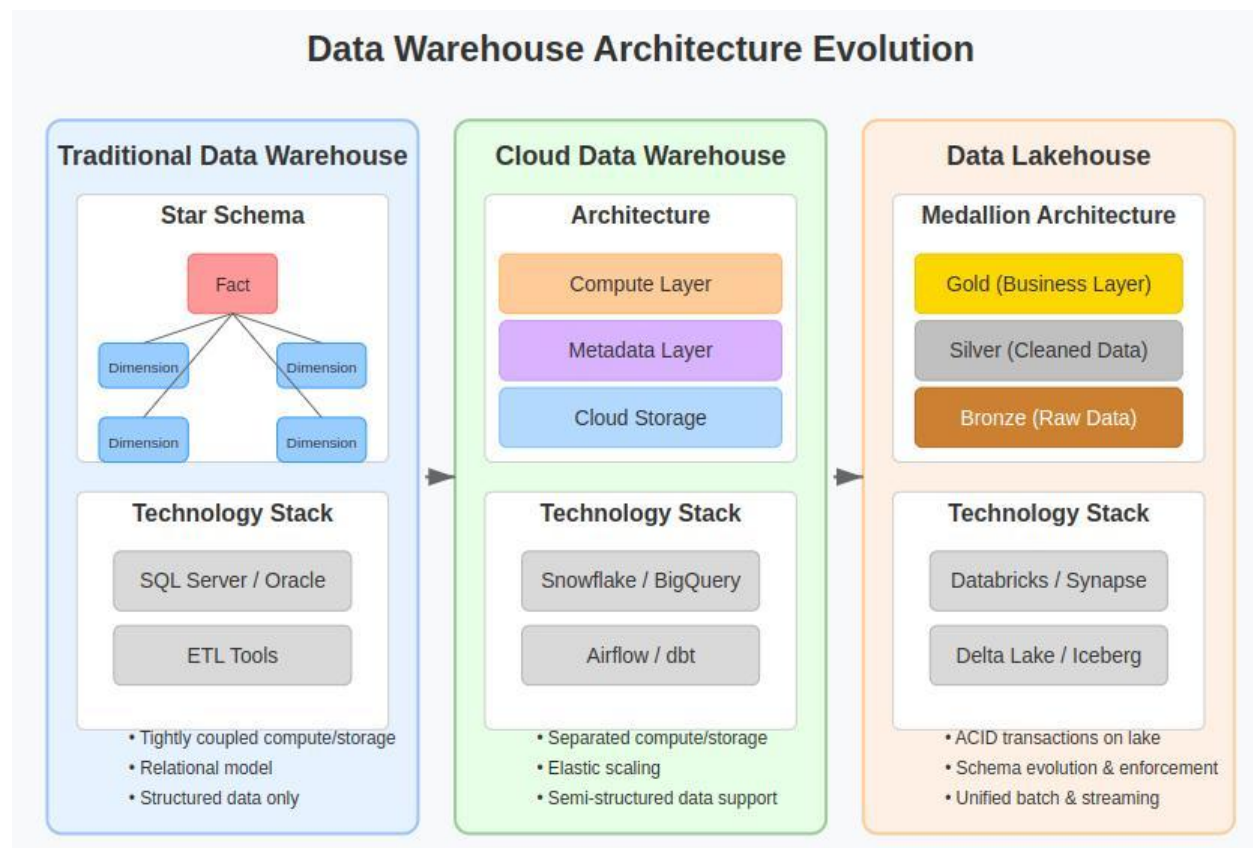
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1. Introduction

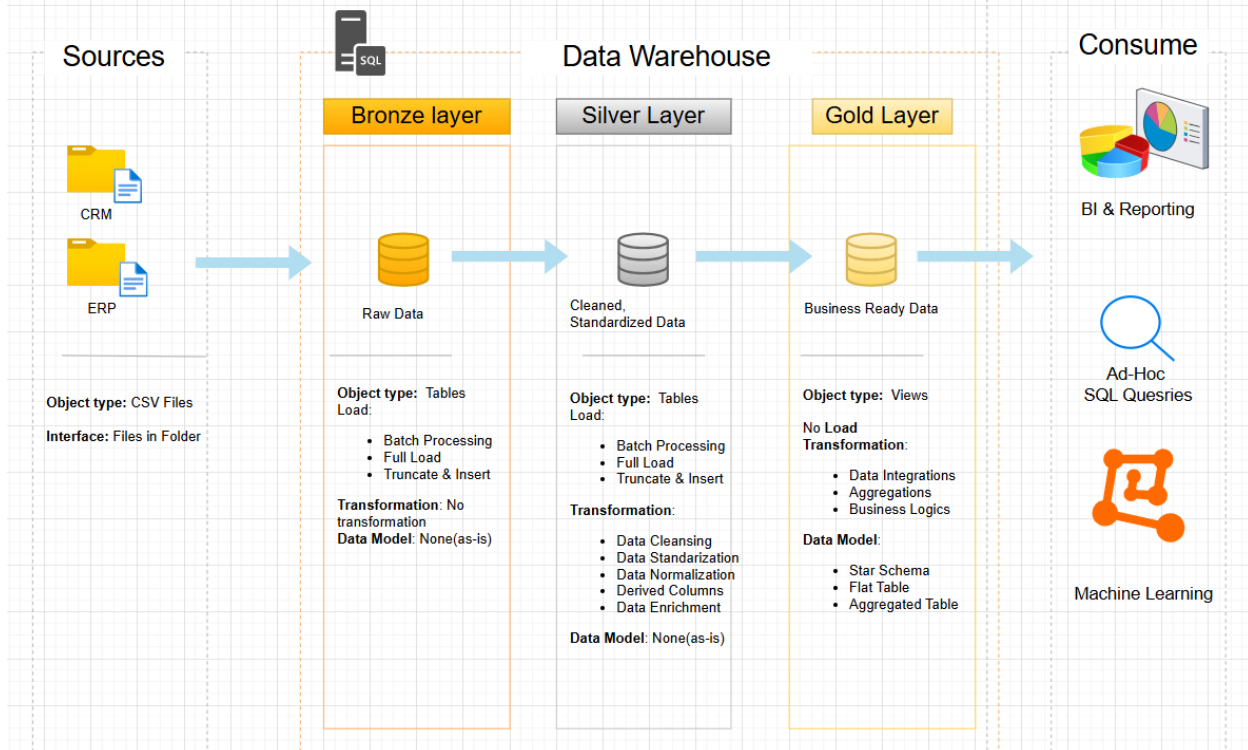
A Data Warehouse is a centralized system designed to collect, store, and analyze data from multiple sources. It supports business intelligence (BI), reporting, and decision-making by transforming raw data into meaningful insights.

This report presents a complete overview of a modern Data Warehouse architecture, including data layers, ETL processes, data modeling, and analytical structures such as the Sales Data Mart.

2. Data Warehouse Architecture Evolution and Choice Made



High Level Architecture



The Data Warehouse follows a **multi-layer architecture**:

2.1 Bronze Layer (Raw Data)

- Stores raw data as received from source systems (CRM, ERP, etc.)
- No transformation is applied
- Used for traceability and debugging
- Loaded using full load (truncate & insert)

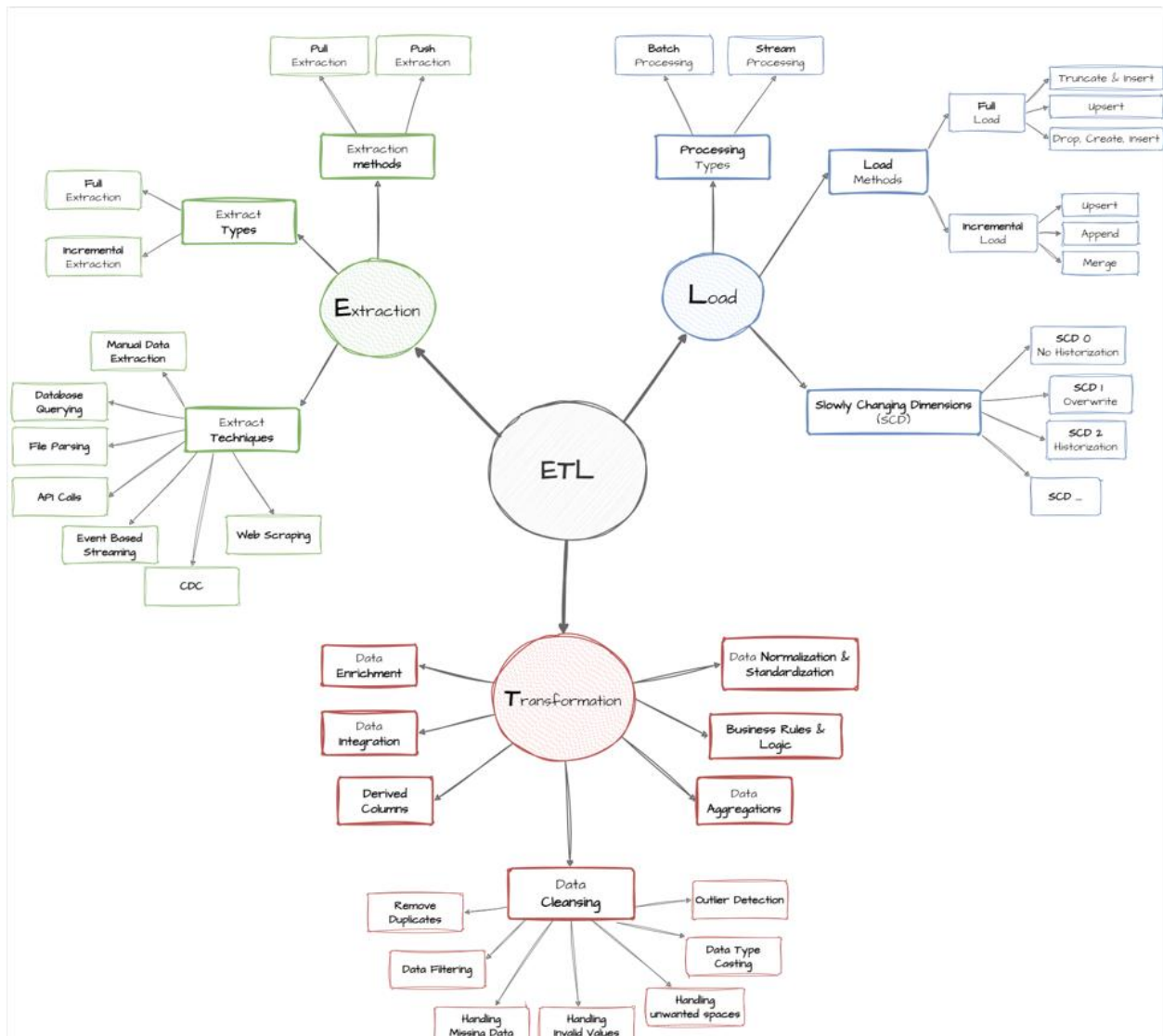
2.2 Silver Layer (Cleaned Data)

- Data is cleaned, standardized, and normalized
- Handles missing values and duplicates
- Prepares data for transformation

2.3 Gold Layer (Business Data)

- Contains business-ready data
- Organized into **fact and dimension tables**
- Used for reporting and analytics

3. ETL Process (Extract, Transform, Load)



The ETL process is essential for moving and transforming data:

3.1 Extract

- Data is collected from multiple sources (CRM, ERP, APIs)

3.2 Transform

- Data cleansing (remove duplicates, fix errors)
- Standardization and normalization
- Data enrichment and integration

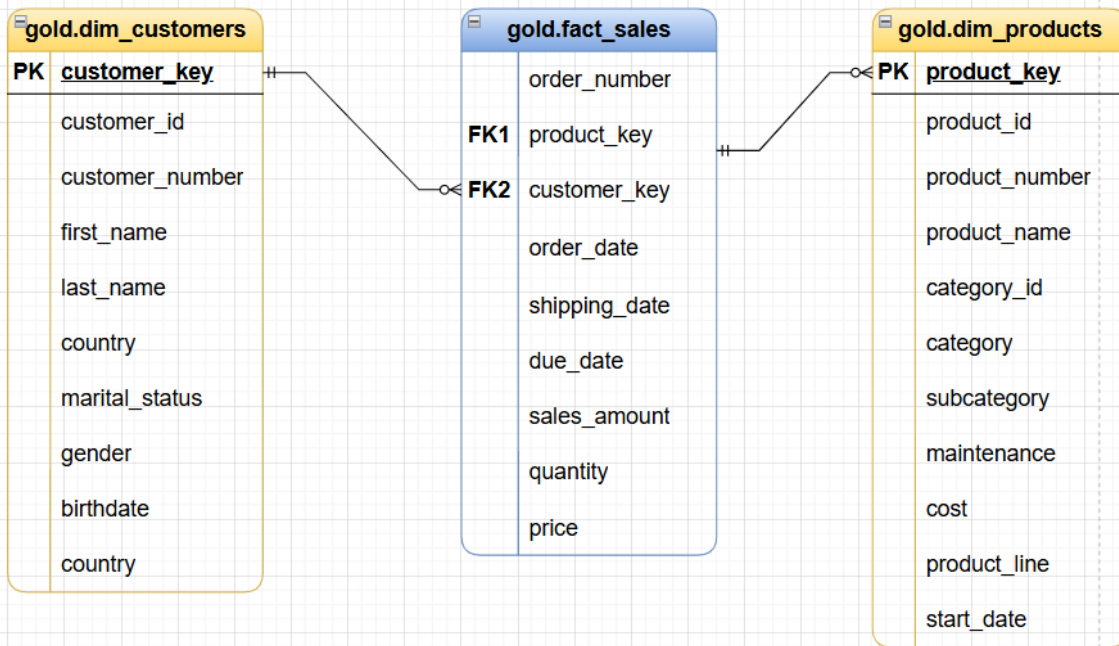
3.3 Load

- Data is loaded into the warehouse layers
 - Methods include:
 - Full Load (truncate & insert)
 - Incremental Load (upsert)
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4. Data Modeling – Star Schema (Sales Data Mart)



DATA MART (Star Schema)



Data Mart Star Schema



The structure of the Data Marts need to be a simple **star schema**, made up of only facts and dimensions.

Simple to manage huge amount of data quickly

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The **Sales Data Mart** is designed using a **Star Schema**, which consists of:

4.1 Fact Table: fact_sales

Stores measurable business data:

- Sales amount
- Quantity
- Price
- Order dates

4.2 Dimension Tables

Provide descriptive context:

- **dim_customers**
 - Customer demographics (name, gender, country, etc.)
- **dim_products**
 - Product details (category, cost, product line, etc.)

4.3 Relationships

- fact_sales.customer_key → dim_customers.customer_key
- fact_sales.product_key → dim_products.product_key

This structure enables fast analytical queries and efficient reporting.

5. Sales Calculation (Business Logic)

The key business metric in this model is **Sales Amount**, calculated as:

Sales = Quantity × Price

This formula is implemented in the transformation logic within the Gold layer.

6. Data Catalog (Gold Layer)

The Gold Layer contains well-defined tables:

6.1 dim_customers

- Stores customer attributes such as name, gender, and country

6.2 dim_products

- Stores product information including category and cost

6.3 fact_sales

- Stores transactional sales data (orders, quantity, price)

These tables are optimized for analytics and reporting.

7. Naming Conventions

To ensure consistency and maintainability:

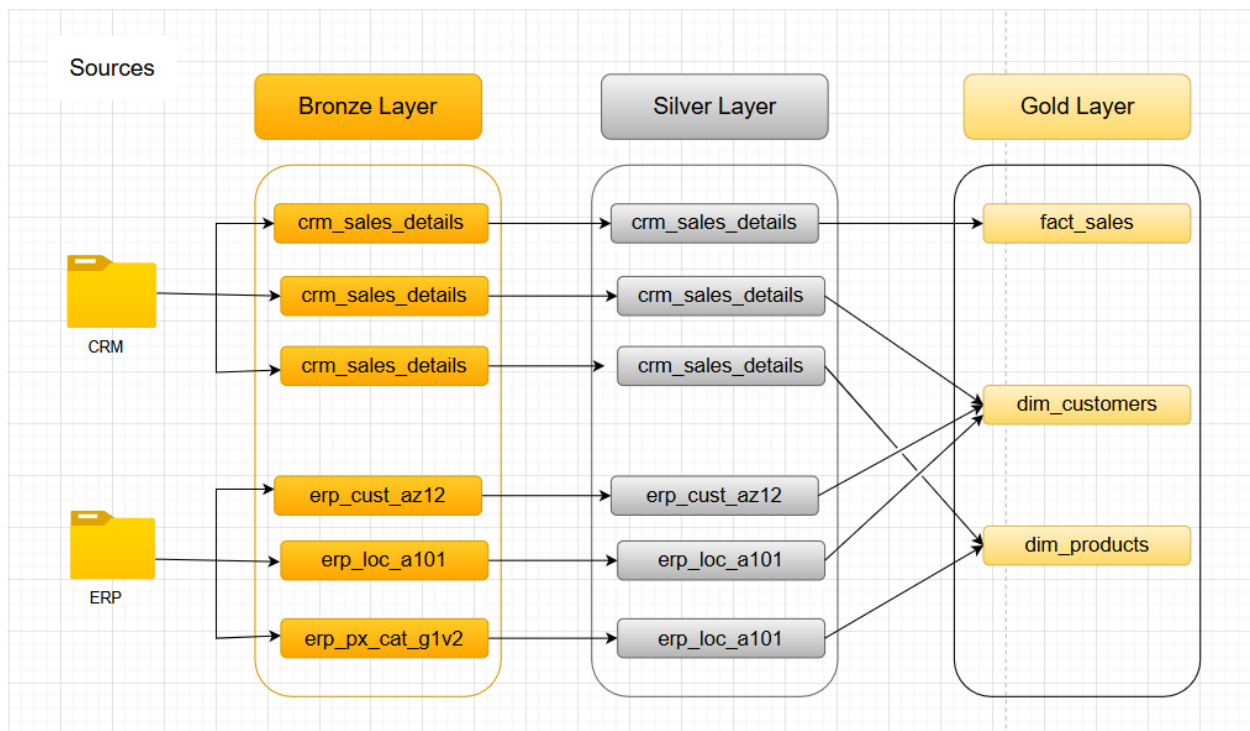
- Use **snake_case naming** (lowercase with underscores)
- Gold layer tables follow:
 - dim_ for dimension tables
 - fact_ for fact tables

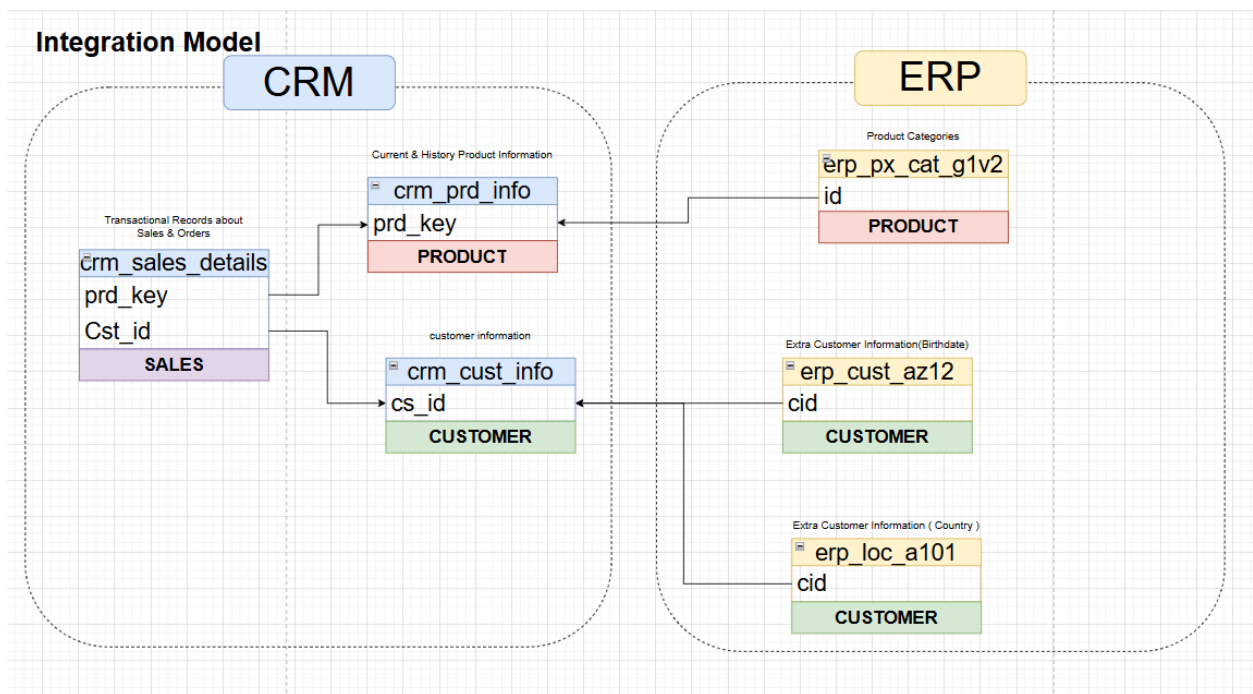
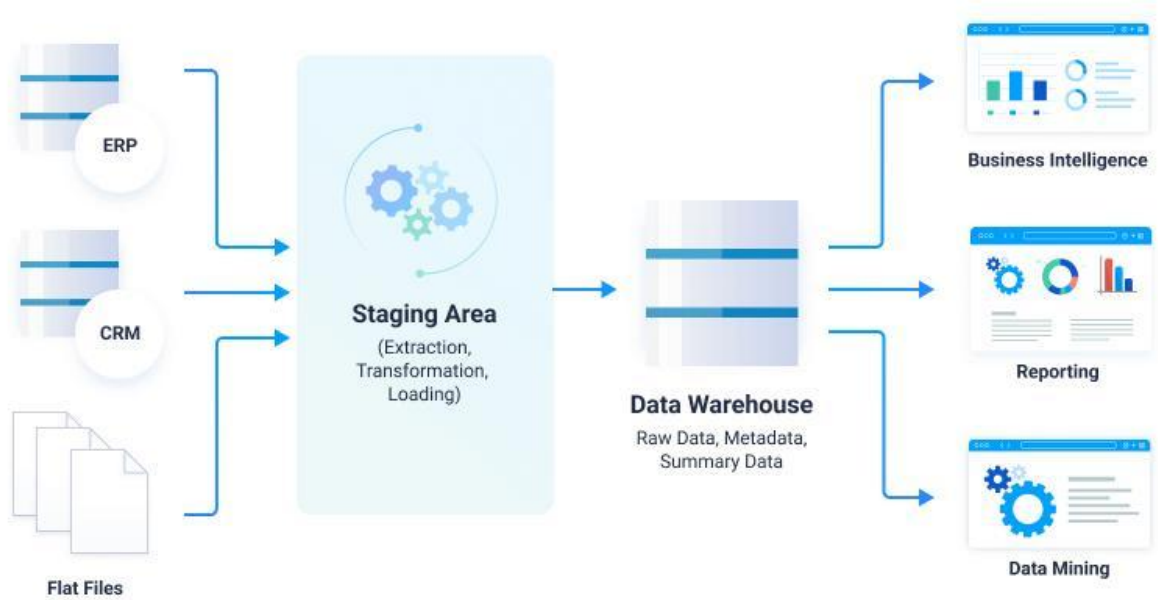
Examples:

- dim_customers
- dim_products
- fact_sales

Surrogate keys use _key suffix (e.g., customer_key, product_key).

8. Data Flow & Integration





Data flows through the system as follows:

1. Source systems (CRM, ERP) generate raw data
 2. Data is ingested into the Bronze layer
 3. Cleaned and transformed in the Silver layer
 4. Aggregated and modeled in the Gold layer
 5. Delivered to BI tools for reporting
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9. Benefits of the Data Warehouse

- Centralized and consistent data source
 - Improved data quality and reliability
 - Faster reporting and analytics
 - Historical data tracking
 - Better decision-making support
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10. Conclusion

This Data Warehouse project demonstrates a complete pipeline from raw data ingestion to business-ready analytics. By implementing layered architecture, ETL processes, and a star schema model, the system ensures scalable, reliable, and efficient data management.

The Sales Data Mart provides a powerful analytical foundation, enabling organizations to derive insights from customer behavior and product performance.

11. References

- Data Catalog Documentation
- Data Layers Architecture
- Naming Conventions

Are Provided in The Git repository:

<https://github.com/Hermann1412/sql-data-warehouse-project/tree/main/docs>
